



Assessment of the Utilization of Cassava Processing Technologies among Women Processors in Ekiti State, Nigeria

* Adewumi, O.T., Ajala, A.O., Bamigboye, O.T, Fanu, A.T, Oke, O.S

Department of Agricultural Economics and Extension, Faculty of Agriculture, Federal University, Oye Ekiti

Correspondence author: Adwumi, O.T oluranti.adewumi@fuoye.edu.ng +2348069451972

ARTICLE INFO.

ABSTRACT

Key words:

Technology Utilization,
Women Cassava Processor,
Cassava Processing
Technologies

The study assessed the Utilization of cassava processing technologies among women processors in Ekiti State, Nigeria. A total of 220 women were sampled from the three Agricultural Development Programme zones in Ekiti State based on the preponderance of women cassava processor. A structured interview schedule was used in the collection of primary data for the research. Pearson Product Moment Correlation analysis was used to test the hypotheses. The result indicated that the mean age of the women was 48.90 years while the standard deviation was 1.01, majority (80%) of the respondents were married with 4.79 mean household size, majority (52.8%) of the cassava processors had secondary education. Availability of cassava processing technologies to rural women revealed that screw press ($\bar{x}=3.71$), milling machine ($\bar{x}=2.89$), grinder ($\bar{x}=2.71$) had high availability, while, fermentation tank ($\bar{x}=1.16$), mechanical fryer ($\bar{x}=1.10$) and peeler ($\bar{x}=1.09$) had low availability. Screw press was most Utilized while fermentation tank, mechanical fryer and peeler were the least Utilized processing technologies. The level of Utilization of cassava processing technologies was low (41.4%). The study concluded that the level of utilization of cassava processing technologies was low. It was recommended that cassava processing plants should be provided by government at subsidized cost for rural women who are into cassava processing to enhance improved Utilization of processing technologies and hence improved their standard of living.

Introduction

Cassava (*Manihot esculenta* or *Manihot utilissima*) is a root vegetable that originated in Brazil and was brought to West Africa by the Portuguese. Cassava is the world's 19th most popular crop, with total output of 278 million tonnes (Food and Agricultural Organisation, FAO, 2012; 2018). It is often recognized as the most productive crop and dietary energy source in the tropics. Fifteen nations produce more than one million tons per year; the top producers are Brazil, Indonesia, Nigeria, Thailand, and Zaire, which contribute for 63 percent of global output (Okpeke *et al.*, 2015). With an annual output of about 34 million tons of tuberous roots, cassava is one of the world's most important food crops (Asogwa *et al.*, 2013).

International Institute of Tropical Agriculture, IITA has played a leading role in developing improved cassava varieties that are disease, pest and drought resistant, low cyanide, early maturing and of high yield. IITA (2020), branded names of top performing improved varieties of cassava for easy identification, cultivation, marketing by farmers and gave the local names. The released varieties with their local names were: IBA961632 (farmer's pride), IBA980581 (Dixon), CR36-5 (Ayaya), IBA070593 (Sunshine), IBA980505 (Fine face), TME419, TMS13F1160P0004 (Game changer), TMS13F1343P0022 (Obasanjo-2), NR130124 (Hope), and TMEB693 (Poundable). There are about seventeen forms into which cassava may be processed in Africa (Kaine, 2018) and the forms into which cassava is processed and consumed is said to be dependent on cultural food habits, tastes and preferences of the people. Some cassava products in Africa included *Chikwangue* (Central Africa), *Ntuka* (Zaire), *Gari* (West Africa), *Attieke* (Ivory Coast); *Fufu* (Nigeria, Ghana and Zaire) These varieties of products have made cassava to be either a primary or secondary staple through the forest and transition zones of Africa (IITA, 2020).

According to FAO (2018), cassava with an annual production of roughly 60 million tonnes in 2018, almost 19% of total world production (Okonkwo, *et al.*, 2020) is an essential food crop produced in large quantities in Nigeria. Cassava is a popular staple in Nigerian families across all cultures and social classes. Cassava per capita consumption is very high and provides about 80% of the total energy intake of many Nigerians (Ani, 2017). As a crop whose by-products have a wide range of uses, cassava is the most important food crop for Nigeria by production quantity next to yam, which is the most important food crop by value (Food and Agriculture Organisation Corporate Statistical Database, FAOSTAT, 2016).

Cassava has been criticized in numerous ways: that it is women's crop (Forsythe, *et al.*, 2015), as they play a crucial role in its production, processing (Walker, *et al.*, 2014) and marketing, however, it depletes soil nutrients, consumed only by poor households, considered a lethal and nutritionally deficient food (Inyada, 2014). These stigmas are not entirely true. The collaborative study of cassava has revealed that both men and women are involved in cassava production, processing and marketing (Olaosebikan *et al.*, 2019). Soils that have been under continuous cassava cultivation for at least ten years were found to be as fertile as soils of other crops (Biratu *et al.*, 2019). Cases of cyanide poisoning from the consumption of cassava are rare. A study by Amai *et al.*, (2021) identified that, the cynogenic glucoside content in cassava are reduced or even eliminated by proper processing especially during fermentation.

Processing, according to Adiaha (2017), entails physical or chemical alteration, storage, packaging, and distribution. The degree of change and the nature of the processing might range from apple cleaning, grading, and packing to milling, cooking, mixing, and chemical alteration that results in textured vegetable snack food. To boost storability, there is need to get a more conveniently or economically transportable form, and improve palatability, nutritional content, and customer convenience.

A major hindrance to the utilization of cassava processing technologies could be either as a result of lack of access or non-affordability of the processing technologies (Adesakin and Oluwatusin 2017). Current research emphasis should therefore be carried out to assess the utilization of cassava processing technologies among rural women. Aside institutional considerations, lack of access to cassava processing technologies could be a major socio-economic factor.

This scenario is witnessed in the underutilization of cassava processing technologies with consequent effect on output and income of cassava processors. In view of this, research institutes such as National Centre for Agricultural Mechanization (NCAM), Product Development Agency (PRODA), International Institute for Tropical Agriculture, (IITA), National Roots Crops Research Institute (NRCRI), Agricultural Engineering Departments in Several Universities and Polytechnics in Nigeria have developed many mechanized units to fabricate processing machines designed to ease cassava processing. The peeling machine, cassava chopping machine, grafting machine, hammer mill, hydraulic press, dryer, and pelletizer are among the models and varieties of cassava processing technologies available on the market (Fefa *et al.*, 2020), despite this, utilization of cassava processing technology among rural women is still low. This study objective is to assess the utilization of cassava processing technologies among rural women while identifying the cassava processing technologies available, accessible and affordable to rural women.

METHODOLOGY

Study Area

The research was carried out in Ekiti State which is located in Nigeria's Southwest region, it is located approximately between latitudes 7.5°N and 8.0°N and between longitudes 4.5°E and 5.0°E. According to the 2006 population census, the state has a population of 2,210,957 people and a land area of 6,352KM². The climate in Ekiti State is tropical, with two distinct seasons of rainy season (April-October) and dry season (November-March). Ekiti State is bordered on the North by Kwara State, on the South by Ondo State, on the West by Osun State and on the East by Kogi State. The land is rich in agricultural resources, with cocoa being the most important cash crop. Food crops like yam and cassava as well as grains like rice and maize are widely cultivated. The study adopted a multi-stage sampling procedure in the selection of respondents. In the first stage, three Local Government Areas were selected randomly from the State based on the preponderance of cassava processing. The second stage involved a random selection of three rural communities from each of the Local Government Areas summing up to nine communities. The third stage involved a random selection of cassava processors from each of the communities making total of 220 cassava processors. Data were collected from both primary and secondary sources. Interview schedule was used in the collection of primary data for the research. Secondary data was obtained from the state ADP office, relevant journals, textbooks and the internet. The dependent variable is utilization of the cassava processing technologies which is the degree to which the processing technologies is being utilized. The level of utilization was determined by making a list of processing technologies of cassava which include 'peeler, grater, screw press, granulator, dryer, fryer, sifter, fermentation tank, milling machine and grinder,' and respondents were asked to indicate the level of utilization of each of the listed methods along a 4 point Likert scale developed by Rensis Likert with values '4- highly utilized', '3- moderately utilized', '2- slightly utilized', '1- not utilized' respectively. The benchmark was calculated as 2.5, while, variables with mean scores less than, within and more than 2.5 was regarded as low, medium and high level of utilization of cassava processing machines respectively. The independent variables were the characteristics of improved technologies- availability, accessibility and affordability. The respondents were given a four-point Likert's scale to rate the technologies thus: highly available, (4 points), moderately available, (3 points), slightly available (2 points), not available (1 point). Highly accessible, (4 points), moderately accessible, (3 points), slightly accessible (2 points), not accessible (1 point) and highly affordable, (4 points), moderately affordable, (3 points),

slightly affordable (2 points), not affordable (1 point). Possible maximum scores for each respondent were 44, while the minimum was 11.

Data were described through frequency counts, percentages, mean score. Pearson Product Moment Correlation (PPMC) analysis was used to show the association between selected socio-economic characteristics of and level of utilization of processing technologies; association between level of utilization of cassava processing technologies and characteristics of the technologies namely: availability, accessibility, affordability etc.

RESULTS AND DISCUSSION

The mean age of women cassava processors was 49 years and 80.0% were married. Result in Table 1 further shows that 74.1% of rural women were involved in part time processing. This agrees with the findings of Okpeke *et al.*, (2015), who reported that majority (66%) of respondent in Isoko North Local government Area of Delta State were into cassava processing on part-time basis. The findings is in line with Adekunle *et al.*, (2018), who reported that majority (81.67%) of the cassava processors were members of cooperative societies in Oyo State. Contrarily, Madu *et al.*, (2018) reported that women involvement in cooperative society is low. Findings revealed that 82.3% of the rural women do not have any of the cassava processing technologies. This finding is in contrast with the finding of Awoyemi *et al.*, (2020) who stated that most of cassava processing technologies were available to majority of the processors in the study area. The Focus Group Discussion (FGD) revealed that most of the women hire cassava processing machine whenever the need arises. The result reveals that 47.7% of the cassava processors sourced their processing capital from cooperative society. Below average (47.3%) of the cassava processors earned less than ₦151,000 average monthly income from cassava processing. The mean annual extension contact was 1.8. This implies that 61.5 percent of the cassava processors had low or no contacts with extension; this could be a reason for the poor access to cassava processing technologies and the poor knowledge of the technologies. Good extension programs and contact with processors are a key aspect in technology dissemination and utilization. Bamigboye *et al.*, (2019) reported that 74.5% of rural women have inadequate extension contact.

Table 1: Socio-economic characteristics of rural women processors

Variable	Frequency	Percentage	Mean
Age (Years)			
40 and below	52	23.6	49
41-50	77	35.0	
51-60	57	25.9	
61 and above	34	15.5	
Marital status			

Single	12	5.5
Married	176	80.0
Widowed	28	12.7
Divorced	4	1.8

Time of involvement in processing

Part time	163	74.1
Full time	57	25.9

Membership of cooperative

Yes	150	68.2
No	70	31.8

Ownership of processing technologies

Yes	39	17.7
No	181	82.3

Source of fund

Personal savings	27	12.3
Relatives and friends	37	16.8
Cooperative societies	105	47.7
Village money lenders	8	3.6
No access to credit	43	19.5

Average Monthly Income (₦)

<₦151,000	104	47.3	
₦151,000-	49	22.3	177,859
₦200,000	30	13.6	
₦201,000-	37	16.8	
₦250,000			

>N251,000

**Extension
contact**

≤ 3	85	38.6	1.80
4-5	55	25.0	
6 and above	11	5.0	
No contact	69	31.4	

1\$=N414.50

Source: Field Survey, 2021

Result in Table 2 shows the rank-order of availability of cassava processing technologies to rural women in descending order. Screw press ($\bar{x}$ =3.71) ranked first followed by milling machine ($\bar{x}$ =2.89), grinder ($\bar{x}$ =2.71), mechanical grater ($\bar{x}$ =2.22), granulator ($\bar{x}$ =2.12), dryer ($\bar{x}$ =1.93), mechanical sifter ($\bar{x}$ =1.59), mechanical dewatering ($\bar{x}$ =1.22), fermentation tank ($\bar{x}$ =1.16), mechanical fryer ($\bar{x}$ =1.10) and peeler ($\bar{x}$ =1.09). It could therefore be deduced from that screw press, milling machine, grinder have high availability. On the other hand, mechanical grater, granulator, dryer, mechanical sifter, mechanical dewatering, fermentation tank, mechanical fryer and peeler had low availability to the respondent. This finding agreed with the study of Awoyemi *et al.*, (2020) that pressing machine and grating machine were highly available, while fryer and dryer were less available for cassava processing. FGD revealed that the processing technology that is highly available in the area is screw press, they attribute the level of availability of screw press to the fact that they are simple, easy to use and required in almost all processing chain of cassava.

Table 2: Rank-order of cassava processing technologies based on their availability.

Cassava Processing Technologies	Mean	Rank
Screw press	3.71	1 st
Milling machine	2.89	2 nd
Grinder	2.71	3 rd
Mechanical Grater	2.22	4 th
Granulator	2.12	5 th
Dryer	1.93	6 th
Mechanical Sifter	1.59	7 th
Mechanical Dewatering	1.22	8 th
Fermentation Tank	1.16	9 th

Mechanical Fryer	1.10	10 th
Peeler	1.09	11 th

Source: Field Survey, 2021

Result in Table 3 shows the rank-order of the Accessibility of cassava processing technologies in descending order. Screw press ($\bar{x}$ =3.90) ranked first followed by grinder ($\bar{x}$ =3.85), milling machine ($\bar{x}$ =3.68), mechanical sifter ($\bar{x}$ =3.61), granulator ($\bar{x}$ =3.33), mechanical grater ($\bar{x}$ =3.23), dryer (c2.28), peeler ($\bar{x}$ =1.15), fermentation tank ($\bar{x}$ =1.11), mechanical fryer ($\bar{x}$ =1.08) and mechanical dewatering ($\bar{x}$ =1.06) which ranked last on the list. This result shows that screw press, grinder, milling machine, mechanical sifter, granulator and mechanical grater were highly accessible, this could be as a result of the proximity of the processing technologies to the processors. On the other hand, dryer ($\bar{x}$ =2.28), peeler ($\bar{x}$ =1.15), fermentation tank ($\bar{x}$ =1.11), mechanical fryer ($\bar{x}$ =1.08) and mechanical dewatering ($\bar{x}$ =1.06) had low accessibility. This could be that dryer, peeler, fermentation tank, mechanical fryer and mechanical dewatering were not within the reach of the processors as some of the rural women had to travel far to access some of the technologies. The report of this finding agreed with Fatuase *et al.*, (2019) who reported that many of the processors in Ekiti State were not aware of the technologies, while some were not available, accessible and affordable to the processors.

Table 3: Rank-order of cassava processing technologies based on their accessibility.

Cassava Processing Technologies	Mean	Rank
Screw press	3.90	1 st
Grinder	3.85	2 nd
Milling machine	3.68	3 rd
Mechanical Sifter	3.61	4 th
Granulator	3.33	5 th
Mechanical Grater	3.23	6 th
Dryer	2.28	7 th
Peeler	1.15	8 th
Fermentation Tank	1.11	9 th
Mechanical Fryer	1.08	10 th
Mechanical Dewatering	1.06	11 th

Source: Field Survey, 2021

Result in Table 4 shows the rank-order of the Affordability of cassava processing technologies to rural women in descending order. Screw press ($\bar{x}$ =3.35) ranked first followed by mechanical grater ($\bar{x}$ =3.30), granulator ($\bar{x}$ =3.15), milling machine ($\bar{x}$ =2.88), grinder ($\bar{x}$ =2.87), mechanical sifter ($\bar{x}$ =2.36), dryer ($\bar{x}$ =2.17), mechanical dewatering ($\bar{x}$ =1.17), mechanical fryer ($\bar{x}$ =1.13), peeler ($\bar{x}$ =1.11), and fermentation tank ($\bar{x}$ =1.07) which ranked last on the list. This analysis shows that screw press, mechanical grater, granulator, milling machine, and grinder were affordable. On the other hand, mechanical

sifter ($\bar{x}$ =2.36), dryer ($\bar{x}$ =2.17), mechanical dewatering ($\bar{x}$ =1.17), mechanical fryer ($\bar{x}$ =1.13), peeler ($\bar{x}$ =1.11), and fermentation tank ($\bar{x}$ =1.07) had low affordability. FGD revealed that some of these women processors were limited financially to acquire these technologies.

Table 4: Rank-order of cassava processing technologies based on their affordability.

Cassava Processing Technologies	Mean	Rank
Screw press	3.35	1 st
Mechanical Grater	3.30	2 nd
Granulator	3.15	3 rd
Milling machine	2.88	4 th
Mechanical Sifter	2.87	5 th
Grinder	2.36	6 th
Dryer	2.17	7 th
Mechanical Dewatering	1.17	8 th
Mechanical Fryer	1.13	9 th
Peeler	1.11	10 th
Fermentation Tank	1.07	11 th

Source: Field Survey, 2021

Result in Table 5 shows that 34.4 percent of the processing technologies had high level of Utilization, 24.3 percent had medium level of Utilization, while 41.4 percent had low level of Utilization. This analysis shows that the level of Utilization of the cassava processing technologies by rural women were low due to their low level of availability in the study area. It also shows that these women processors had low access to them and were not able to afford them. This finding agreed with Ewebiyi *et al.*, (2020), who reported low level of Utilization among majority (56.8%) of the cassava processors in Oyo State. FGD conducted revealed that some women had Utilized some of the cassava processing technologies in the pass, but due to the fact that they could not afford it, they had to revert to the traditional method of processing cassava. Some of them said cassava peeler for instance was not too efficient as they claimed that there is usually wastage of the valuable flesh of cassava in the process of peeling.

The result was in contrast with Ehinmowo *et al.*, (2016), who reported a high level of adoption (67%) for the cassava processing technologies in South West, Nigeria. Likewise, Fatuase *et al.*, (2019), reported high level of adoption (67%) of the cassava processing technologies in Ekiti State. Also, Adisa *et al.*, (2013) reported high level of adoption (76%) of cassava processing technologies by rural women in Irepodun Local Government Area of Kwara State. In summary, it could be said that the cassava processing technologies available in the study area had low level of Utilization.

Table 5: Distribution of women cassava processors based on the level of Utilization of cassava processing technologies

Level	Scores	Frequency	Percentage
High	Above 33	76	34.5
Medium	22-32	53	24.1
Low	Below 22	91	41.4
Total		220	100

Source: Field Survey, 2021

Test of Hypotheses

H1: There is no significant relationship between the level of utilization of cassava processing technologies and selected women processors' socio-economic characteristics, namely; years of processing experience, level of involvement in processing, membership of cooperative society, source of fund and contact with extension.

Data in Table 6 shows the result of correlation analysis of the variables. A positive and significant correlations was found between level of Utilization of processing technologies and source of fund ($r = 0.254$). Coefficient of determination, r^2 , in Table 6 shows the percentage variation in Y variable (level of utilization of processing technologies) as explained by each of the X variables (women processors' socio-economic characteristics) in the study. 6.4 percent variation in level of Utilization was attributed to the source of fund. This result implies that the more the sources of fund and funds available to women cassava processors, the higher the level of Utilization of the cassava processing technologies. This may be true if the funds available to women cassava processors will make them purchase the processing technologies or if it will increase their level of Utilization of the processing technologies. This finding is not consistent with Awoyemi *et al.*, (2020) who reported that educational level and cassava processing experience was significantly related to the usage of improved cassava processing technologies. Based on the result presented in the table below, the null hypothesis was rejected.

Table 6: Correlation between the level of utilization of cassava processing technologies and selected women processors' socio-economic characteristics

Variable	R	r ²	Percentage contribution %	Decision
Years of experience	-0.033	0.001089	0	NS
Level of involvement in processing	-0.032	0.001024	0	NS
Membership of cooperative	-0.067	0.004489	0	NS
Source of fund	0.254**	0.064516	6.4	S
Extension contacts	0.033	0.001089	0	NS

** Significant at 0.01 level of significance

NS-Not significant

S-Significant

r- Pearson Correlation Coefficient

r²-Coefficient of Determination

Table 7: Summary of the relationship between the level of Utilization of cassava processing technologies and the characteristics of the processing technologies.–

Variable	R	r ²	Percentage contribution %	Decision
Availability	0.311**	0.096721	9.6	S
Accessibility	0.141*	0.019881	1.9	S
Affordability	0.76	0.5776	5.7	NS

*Significant at 0.05 level of significance

** Significant at 0.01 level of significance

NS-Not significant

S-Significant

r- Pearson Correlation Coefficient

r²- Coefficient of Determination

Hypothesis Two: There is no significant relationship between the level of Utilization of cassava processing technologies and its availability, accessibility and affordability.

The correlation analysis in Table 7 shows positive and significant relationship between level of Utilization of cassava processing technologies among rural women and its availability, $r = 0.311$, at 0.01 level of significance and accessibility, $r = 0.141$, at 0.05 level of significance. The coefficient of determination (r^2) shows that 9.6 percent and 1.9 percent variation was attributed to rural women

availability, and accessibility to cassava processing technologies respectively. This implies that availability had influence on the level of utilization of cassava processing technologies. This may be true if processors will utilize the cassava processing technologies that are available to them. In the same vein, accessibility was related to the level of utilization of the cassava processing technologies. This may be true if the women processors take advantage of the few processing technologies that are accessible to them, and if the processing technologies are relevant to the need of the processors.

CONCLUSION AND RECOMMENDATIONS

The level of utilization of cassava processing technologies was low in the study area to enhance their standard of living. Screw press is the most utilized cassava processing technologies by the processors while fermentation tank is the least Utilized processing technologies. On the bases of availability, screw press is the most available, accessible and affordable processing technologies in the study area. Source of fund of women processor had

positive and significant relationship with the level of utilization of cassava processing technologies. Availability and accessibility of cassava processing technologies had positive and significant relationship with the level of utilization of processing technologies by rural women. The study recommends that cassava processing plants should be provided by government at subsidized cost for rural women who are into cassava processing. Also, governments should provide necessary support for the commercialization of feasible cassava processing machinery developed by research organizations, which if produced in large quantity will be available to the processors at a subsidized rate. Lastly, women should be sensitized and encouraged on the need to adopt and utilise the improved cassava processing technologies for increased and improved output.

REFERENCES

- Adekunle, A Adeoye, A, Oyeleye, A.A, Ajibola, A.J, Daud, S.A, Alabi, A.F and Amao, S.A
- (2018). Women Participation in Cassava Processing in Ibarapa Central Local Government Area of Oyo State, Nigeria. *International Journal for Research in Agricultural and Food Science*. 4(4), 15-28
- Adesakin, M.F and Oluwatusin, F.M (2017), Profitability Analysis of the Adoption of Improved Agricultural Technologies among Cassava Farmers in Ekiti State, Nigeria. *Journal of American Science*, 13(4), 115-121
- Adiaha, M.S. (2017), Economics of Maize (*Zea mays* L) Production in Nigeria and Maize Traditional Utilization.

- International Journal for Scientific World* 5(2), 106-109
- Adisa, R.S, Olatinwo K.B and Shola-Adido, O, (2013). Adoption of Cassava Processing Innovations among Rural Women in Irepodun Local Government, Area, Kwara State, Nigeria. *Journal of Production Agriculture and Technology*. 9(1), 1-12
- Amai, I.U, Ogonbna, I.O, Aguru, C.U and Amai, D.C. (2021). Aflatoxin contamination of Gari sold in some selected markets in Benue State. North Central. Nigeria. *Advances in Microbiology*. 11, 499-509
- Ani, S.O and Agbugba, I (2017). Economics of marketing of cassava derivatives (Floor, Garri and Chips) in Uzo-Uwani Local Government Area, Enugu State. *Delta Agriculturists*. 9(1), 46-54
- Asogwa BC, Ezihe JAC, and Ater PI (2013). Socio-economic analysis of cassava marketing in Benue State, Nigeria. *International Journal of Innovation and Applied Studies*, 2(4): 384-391.
- Awoyemi, A.O, Adesokan O.J, Kayode, A.O, Omotesho, K.F and Osasona, K.K. (2020). Assessment of cassava processing technologies usage among rural women in Kwara State, Nigeria. *Cercetari Agronomice in Moldova*, LIII (3): 183.
- Bamigboye O.T., Ogunjimi S.I and Adara C.T (2019). Constraints to activities of women agricultural cooperative societies in Emure Local Government Area of Ekiti State. *Nigerian Journal of Rural Sociology*. Vol. 19(2), pp 15-19
- Biratu, G.K Elias, E and Ntawuruhunga, P. (2019). Soil fertility status of cassava fields treated by integrated application of manure and NPK fertilizer in Zambia. *Environmental System Research*. 8(1)
- Ehinmowo O.O, and Fatuase A.I (2016), Adoption of Improved Cassava Processing Technologies By Women Entrepreneur in South – West, Nigeria, *World Journal of Agricultural Research*, 2016, Vol. 4, No. 4, 109-113
- Ewebiyi, I.O Ikotun, T.O and Olayemi, O.O (2020) Constraints to utilization of improved processing technologies among cassava processors in Oyo State. *Journal of Science and Sustainable Development*. 7(1), 41-49
- Fatuase, L.O, Mogbojuri, A.O and Anyache, C.O. (2019). Adoption of cassava processing technologies among entrepreneurs in Ekiti State, Nigeria. *Journal of Natural Science, Engineering and Technology*. 18(1&2): 143-154
- Fefa, J, Obute, C.O, and Igbabeecherwuhe, S (2020). Cassava Processing Technology Adoption and Poverty Reduction among operators in Benue State. Nigeria. *International Journal of Science and Research (IJSR)*. 3(10), 309-317
- Food and Agricultural Organisation, FAO(2012). Agricultural Statistics, Food and Agricultural Organisation of the United Nation. Rome <https://faostat.fao.org/>
- FAO (2016). Food and Agriculture Organisation of the United Nation Statistical Database –FAOSTAT, <http://faostat.fao.org/>

- FAO (2018). Food Outlook- Biannual Report on Global Food Markets- November 2018. Rome, p.104.<https://www.fao.org/3/ca2320en/CA2320EN.pdf>
- Forsythe, L., Martin, A.M. and Posthumus, H. (2015). Cassava market development: a path to women's empowerment or business as usual? *Food Chain* 5(1-2): 11-27.
- International Institute of Tropical Agriculture, IITA (2020). Nigeria approves new names for top performing cassava varieties. International Institute of Tropical Agriculture, News and Events, 26th September, 2020
- Inyada, A.E (2014). Economics of Processing Cassava into Gari and Pellets in Kogi State, Nigeria, (Unpublished M.Sc Thesis), Department of Agricultural Economics, UNN.
- Kaine, A.I.N, (2018), Economic of Traditional Cassava Processing Technology Among Small-Holder Female Cassava Processors in Delta North Agricultural Zone. Delta State, Nigeria. *Bangladesh Journal of Africultural Economics*. 1&2, 85-92
- Madu, .T.U, Justin Okoro, M, Ikpan, A.S and Okoye, B.C. (2018). Gender preferred traits on Cassava production and processing value chain in Imo State, Nigeria. *Nigerian Agricultural Journal*. 49(1): 152-163.
- Okonkwo, U.C, Okafor, C.E and Iheeze, C.C (2020). Development and Performance of Evaluation of a Double Barrel Cassava Grating Machine. *Sumerians Journal of Scientific Research*. 3(11). 133-140
- Okpeke M. Y. and Onyeagocha S.U.O. (2015). Analysis of Processing Cassava Tubers into Garri in Isoko North Local Government Area of Delta State, Nigeria. *European Journal of Agriculture and Forestry Research*, 3(5) pp 17
- Olaosebikan, O. Abdulrazaq, B, Owoade, D., Ogunade, A., Aina, O., Ilona, P., Muheebwa, A., Teeken, B., Iluebbey, P., Kulako, P., Bakare, M, and Parkes, E. (2019). Gender-based constraints affecting biofortified cassava production, processing and marketing among men and women adopters in Oyo and Benue States, Nigeria. *Physiological and Molecular Plant Pathology*, 105, 17-27
- Walker, T., Alene, A., Ndjeunga, J., Labarta, R., Yigezu, Y., Diagne, A., Andrade, R., Andriatsitohaine, R.M., De Groote, H., Mausach, K., Yirga, C., Simtowe, F., Katungi, E., Jogo, W., Jaleta, M. and Pandey, S. (2014). Measuring the effectiveness of crop improvement research in sub-Saharan Africa from the perspective of varietal output, adoption and change: 20 crops, 30 Countries and 1150 Cultivars in Farmers' Fields. Synthesis Report for Objectives 1 and 2 of Bill & Melinda Gates Foundation's Diffusion and Impact of Improved Varieties in Africa (DIIVA) Project. CGIAR Independent Science and Partnership Council.